The sharp corner threshold for five-colour cubes

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Abstract

Every collection of fifteen cubes, each displaying all five colours of a fixed palette, contains eight that form a $2 \times 2 \times 2$ cube with monochromatic exterior faces using those five colours. Fourteen need not suffice: a counterexample uses just three varieties, with multiplicities $4, 5, 5$. This determines the five-colour corner value listed as unknown by Berkove, Eickemeyer and Schulman (2018). The lower bound has an elementary proof. For the upper bound, anchored Hall deficiencies and a shared-corner argument give 357 finite cases. Containment, budget bounds and a deletion lemma give 279 smaller problems, each with a separately checked interval certificate. Lean proves the interval and packed checkers sound, verifies all 279 certificates and the fourteen-cube obstruction, and connects the finite problems to physical cube arrangements, yielding a complete formal proof of the sharp threshold.

AI transparency. The research, code and manuscript were developed using GPT-6-Astra (OpenAI), through Codex, under the author's direction. This working manuscript has not received independent human peer review; Section 6 states the formal verification scope.

1 Introduction

MacMahon's coloured-cube puzzles [1] lead to a family of extremal questions: how many face-coloured cubes guarantee that eight can be chosen to make a larger cube with monochromatic faces? Each small cube here uses all colours of the specified palette. Colours remain fixed, proper rotations identify varieties, and reflections are not allowed as moves. Collections may contain repeated varieties. Only exterior faces are constrained; touching interior faces need not have equal colours.

Berkove, Eickemeyer and Schulman [2, Section 8, Table 7] tabulate the corresponding frame numbers for palettes of two to six colours and leave the five-colour column unknown. Their corner values for two, three, four and six colours are respectively $11, 23, 11, 24$. We determine the missing five-colour value; these thresholds need not increase with the palette size. For a $2 \times 2 \times 2$ cube the frame consists of its eight corners. Let N_5 denote the least integer such that every collection of N_5 five-colour cubes contains a solution using all five colours on the target.

Theorem 1. *The sharp five-colour corner threshold is* $N_5 = 15$.

The conclusion for larger collections follows by choosing any fifteen cubes. The theorem concerns a target that may depend on the collection; it does not assert the ability to construct every target.

The proof has two distinct parts. A short corner argument excludes every target for an explicit fourteen-cube collection. For the upper bound, Hall's theorem gives linear alternatives for failure of a target. We cover hypothetical fifteen-cube counterexamples by 357 cases. Containment reduces these to 308, and initial budget bounds exclude another 29. A deletion lemma reduces the totals and budgets in the remaining 279 problems, which are excluded by finite

interval certificates. The certificates, C++ programs, geometric audits, manuscript and Lean sources are included in the accompanying review archive.[1]

The companion paper [4] considers four-colour sources and targets using exactly three colours. It gives the sharp frame threshold $16n - 3$ for every side length $n \geq 2$, with a classification of the largest failing collections. Here both sources and targets use five colours, and the result concerns the corner case $n = 2$. The developments share the palette-independent geometry in `ColourCubes` [5]; neither threshold theorem is used to prove the other.

A limited literature search has found no later value for the five-colour entry. This is not a comprehensive priority claim. In particular, Rosenfeld's October 2026 article [6] has not been examined in full.

2 Geometry and Hall deficiencies

Represent a cube by its colours on the faces $(+x, -x, +y, -y, +z, -z)$ in a right-handed coordinate frame. A five-colour cube has one repeated colour. There are 75 varieties: for each repeated colour, three have its two faces opposite and twelve have them adjacent. In the opposite case, the other four colours form two opposite pairs, giving three possibilities. In the adjacent case, fix the two equal faces; the 24 orders of the other colours are identified in pairs by the rotation exchanging those equal faces. Thus there are 15 opposite-repeat varieties and 60 adjacent-repeat varieties. Each class is one orbit under colour permutations.

A corner type is the ordered triple of its incident face colours, modulo cyclic permutation. Order the outward face normals positively, equivalently read the triple counterclockwise from outside along the body diagonal. This reverses the clockwise convention in [2], consistently on every cube, and therefore preserves all matching and containment relations. Repeated occurrences of a corner type are retained as distinct target positions.

Lemma 2 (Corner–rotation correspondence)**.** *A source cube fits a target corner by a proper rotation if and only if it has the same cyclic corner type at some corner.*

Proof. A proper rotation preserves the orientation of a corner frame, so it induces a cyclic permutation of its three outward faces. Conversely, any two positively oriented corner frames, with a prescribed cyclic alignment, are related by a determinant-positive signed permutation of the coordinate axes. This is a proper rotation of the cube. □

The palette-independent Lean version of this lemma is reused from the shared `ColourCubes` library [5].

Fix a target T. Form a bipartite graph whose left vertices are its eight corner positions and whose right vertices are the individual source copies. A copy is adjacent to a position when it fits there. A matching covering the eight positions is exactly a solution: each chosen copy can be rotated independently, since interior faces are unrestricted. By Hall's theorem [3], failure is equivalent to a nonempty set D of target positions having fewer than $|D|$ neighbouring copies.

Index the varieties by $i = 1, \ldots, 75$, and let x_i be their multiplicities. For a set S of varieties, write $x(S) = \sum_{i \in S} x_i$. Let $N_T(D)$ be the varieties fitting at least one position of D. Then failure of T is equivalent to the existence of a cut

$$x(N_T(D)) \leq |D| - 1, \qquad \varnothing \neq D \subseteq \{1, \ldots, 8\}. \tag{1}$$

There are 255 cuts per target, hence 19,125 cuts altogether.

[1] Intended repository: `https://github.com/jasondavies/five-colour-cubes`. Public release is pending; the review archive includes source checksums, the base Git revision and bundled pinned project dependencies.

3 An elementary fourteen-cube obstruction

Consider four copies of A, five of B, and five of C, where

$$A = (0, 0, 1, 2, 3, 4), \quad B = (0, 1, 2, 3, 2, 4), \quad C = (0, 3, 1, 2, 4, 1).$$

Their corner types are shown below. Triples denote cyclic classes; the last column counts positions whose type occurs in neither of the other varieties.

Variety	Corner types	Private positions
A	$013, 014, 023, 024, 031, 032, 041, 042$	5
B	$022, 023, 034, 042, 122, 124, 132, 143$	6
C	$011, 014, 021, 042, 113, 123, 134, 243$	6

Lemma 3. *The collection $4A + 5B + 5C$ has no five-colour solution.*

Proof. A solvable target must have every corner type in the displayed union. Its only two-colour types are $022, 122, 011, 113$. An adjacent-repeat target therefore repeats either colour 2, with types $022, 122$, or colour 1, with types $011, 113$. In either case these two corners determine the colours on the faces at the ends of the repeated-colour edge. The two orders of the remaining colours give a mirror pair: B and its mirror, or C and its mirror. The mirror of B requires 043, and the mirror of C requires 012; both types are absent. Only B, C remain.

For an opposite-repeat target, both orientations of each occurring unordered triple must be present. The unordered triples for which both cyclic orientations are available are

$$013, 014, 023, 024, 123, 134.$$

If its repeated colour is r, delete r from the four triples containing it. The remaining pairs must form a $K_{2,2}$ on the other four colours: choose one face from each of their two opposite pairs. Colours 2 and 4 occur in fewer than four listed triples. For $r = 1$ or $r = 3$, the four pairs have degree sequence $(3, 2, 2, 1)$, so do not form $K_{2,2}$. For $r = 0$, the pairs are $13, 14, 23, 24$, forcing opposite pairs $\{1, 2\}$ and $\{3, 4\}$. This is A.

Thus only A, B, C can be targets. Their private positions require respectively at least five, six and six copies of the corresponding variety, exceeding the available four, five and five copies. None is solvable. □

4 Reduction to finite problems

Eight copies of any variety solve that variety, so a solution-free collection has every multiplicity at most seven. Suppose such a collection has size 15.

Lemma 4 (Anchored Hall cover)**.** *Suppose a collection contains m copies of a variety A and does not solve A. There is a set K of its target positions such that*

$$|K| \le 7 - m, \qquad \#\{\textit{other copies fitting some position outside } K\} \le 7 - m - |K|. \tag{2}$$

Conversely, such a set certifies failure of target A.

Proof. Apply (1) to A and put $K = D^c$. Every copy of A is a neighbour of the nonempty set D. Subtract these m copies from the bound $|D| - 1 = 7 - |K|$. The converse is the same Hall inequality read backwards. □

First suppose the maximum multiplicity m is at least two. Choose an adjacent-repeat anchor attaining m if possible; otherwise choose an opposite-repeat anchor attaining m. In the latter case all adjacent-repeat multiplicities are at most $m-1$. Relabel the colours to send the anchor to a fixed representative of its orbit.

For each cover K in Lemma 4, put

$$S(K) = N_A(K^c) \setminus \{A\}, \qquad b(K) = 7 - m - |K|.$$

Write $S = S(K)$ and $b = b(K)$. The case has $x_A = m$, $x(S) \le b$, and the multiplicity bounds just described. Additional upper bounds use two-variety solutions. If a copies of A and $8-a$ copies of V build a target, with $a \le m$, then $x_V \le 7 - a$ in a solution-free collection. These tests are finite: the two varieties must cover all eight target positions, A must fit at least a, and V at least $8-a$. They are also sufficient: at most a positions are unavailable to V, and all of them are available to A. Assign these to A, extend the assignment to a positions that A fits, and assign the remaining $8-a$ positions to V.

Colour permutations and simultaneous reflection of the entire collection preserve solvability. First identify equal pairs $(S(K), b(K))$, even when they arise from inequivalent position subsets K; then retain one representative per orbit of these pairs under the anchor stabiliser. The reported counts are constraint-orbit counts, not position-subset counts. Transporting a cover also transports its constrained varieties, budget and multiplicities, so no counterexample is lost. Global reflection is used only to transport a whole hypothetical counterexample, not as an allowed move on an individual cube. The repeated-variety cases have the following counts, totalling 244.

Anchor / maximum multiplicity	2	3	4	5	6	7
Adjacent-repeat	72	55	31	14	4	1
Opposite-repeat	26	21	11	6	2	1
Total	98	76	42	20	6	2

For distinct collections a different choice of anchor reduces the count.

Lemma 5 (Shared two-colour corner)**.** *Among eleven adjacent-repeat cubes, two share a two-colour corner type. If a distinct collection contains such a pair and one member is chosen as anchor, the anchor's two two-colour positions cannot be Hall-deficient.*

Proof. Each adjacent-repeat cube has exactly two such types, *aab* and *aac*, with $b \ne c$. There are only $5 \cdot 4 = 20$ types *aab*. Eleven cubes contribute 22 incidences, so two cubes share a type. For the two characteristic positions of the chosen anchor, the anchor itself and the other cube are two distinct neighbours. Their neighbourhood therefore has size at least two. □

If an opposite-repeat variety is present, choose it as anchor; this gives 31 constraint orbits with every multiplicity at most one. Otherwise use Lemma 5 to choose an adjacent-repeat anchor. Omit the cover complementary to its two characteristic positions and set all opposite-repeat capacities to zero. Its stabiliser preserves these positions, so the omission is compatible with the orbit reduction. It removes one of the 83 adjacent-repeat constraint orbits, leaving 82 further cases.

Proposition 6. *Every solution-free fifteen-cube collection is carried by colour relabelling and, if needed, global reflection into one of the 357 exported root cases.*

Proof. The preceding arguments cover repeated and distinct collections. The finite catalogue, two-variety bounds and coverage of all anchor-cover orbits are checked from physical rotations by the accompanying audit. It regenerates rotations from quarter turns, separately from the

signed-axis permutations used by the case generator, and checks every exported bound and every cover orbit. The resulting counts are

$$244 + 31 + 82 = 357.$$

The cases need not be disjoint; completeness of the cover is what is required. □

Each root specifies an anchor A of multiplicity m, coordinate bounds ℓ, u, a set S of non-anchor varieties and a budget b. The bounds fix $\ell_A = u_A = m$, with $\ell_i = 0$ for $i \neq A$. A vector $x \in \mathbb{Z}^{75}_{\geq 0}$ is *admissible* for this root if

$$\begin{gathered} \ell \leq x \leq u, \qquad \sum_{i=1}^{75} x_i = 15, \qquad x(S) \leq b, \\ \text{for every target } T, \text{ there is } \varnothing \neq D \subseteq \{1, \ldots, 8\} \text{ with } x(N_T(D)) \leq |D| - 1. \end{gathered} \tag{3}$$

Thus admissibility includes failure of every target, in addition to the anchor constraints.

The cover can be reduced without further search. If $S' \subseteq S$ and $b' \geq b$, then $x(S) \leq b$ implies $x(S') \leq b'$ for nonnegative x. Together with coordinatewise inclusion of the root boxes, this identifies 49 cases contained in other cases, leaving 308. Write u for the root upper bounds. In 29 retained cases, $b + u(S^c) < 15$ already contradicts the prescribed total, so only 279 cases need further exclusion proofs. The exact map and retained case numbers are in `data/q5-root-reduction.json`; Lean checks all containments and arithmetic exclusions. The archive retains all 357 original cases.

Lemma 7 (Hall-budget projection)**.** *Suppose a root has anchor multiplicity m, budget $x(S) \leq b$, and total 15, with the anchor outside S. For $0 \leq s \leq b$, put $G = 15 - b + s$ and suppose $G \geq m$. Every admissible vector x has a vector $y \leq x$ with the same anchor multiplicity, total G, budget $y(S) \leq s$, and a Hall deficiency for every target. It obeys the original upper bounds, tightened to $\min(u_i, s)$ on S.*

Proof. First retain only $\min(x(S), s)$ copies in S. This removes at most $b - s$ copies and leaves the anchor untouched, so the remaining total is at least G. Delete further non-anchor copies until the total is G; this is possible because $G \geq m$. Every coordinate decreases, so all upper bounds and Hall deficiency inequalities are preserved. Nonnegativity and $y(S) \leq s$ give $y_i \leq s$ for $i \in S$. □

For our anchored covers, $b = 7 - m - |K|$, so

$$G = m + 8 + |K| + s.$$

Thus the reduced collection consists of the m fixed anchor copies and $8 + |K| + s$ other copies; in particular, $G \geq m$ is automatic. Apply the lemma after selecting and normalising the original root, retaining its bounds. No new anchor selection is required for the smaller collection. Of the 279 remaining roots, 277 admit an exclusion certificate with $s = 0$. The other two, original cases 68 and 69, use $s = 1$ and $G = 12$. Their $s = 0$, $G = 11$ problems actually have solution-free collections, supplied and checked by physical matching in the projected manifest. The projected totals range from 9 to 15. All 101 roots whose original budget was zero retain their original certificates.

5 Interval certificates

For a projected problem, admissibility is defined by (3) with the tightened upper bounds, total G and budget s. We certify that no admissible vector exists.

Write a box as $\ell \le x \le u$. For a cut $x(S) \le r$, two necessary conditions for feasibility in the box are

$$\ell(S) \le r, \qquad G \le r + u(S^c). \tag{4}$$

The first follows from the lower bounds; the second follows from $G = x(S) + x(S^c)$. When they hold, put $s = r - \ell(S)$ and $g = r + u(S^c) - G$. Every satisfying vector obeys

$$x_i \le \min(u_i, \ell_i + s) \quad (i \in S), \qquad x_i \ge \max(\ell_i, u_i - g) \quad (i \notin S). \tag{5}$$

For example, the upper bound follows by giving all other coordinates of S their lower bounds; the lower bound follows by giving all other coordinates of S^c their upper bounds.

For a target, retain its cuts satisfying (4). If none remain, no vector in the box can fail that target. Otherwise let I and U be the intersection and union of their neighbourhoods, and let s_*, g_* be the largest corresponding s, g. Then all vectors in the box that fail this target remain in the refined box

$$u_i' = \begin{cases} \min(u_i, \ell_i + s_*) & i \in I, \\ u_i & i \notin I, \end{cases} \qquad \ell_i' = \begin{cases} \max(\ell_i, u_i - g_*) & i \notin U, \\ \ell_i & i \in U. \end{cases} \tag{6}$$

Indeed, a failing vector satisfies one of the retained cuts, and its bounds in (5) imply (6). The root budget is treated as a family containing a single cut. The total $\sum_i x_i = G$ gives

$$\ell_i' = \max\left(\ell_i, G - \sum_{j \ne i} u_j\right), \qquad u_i' = \min\left(u_i, G - \sum_{j \ne i} \ell_j\right).$$

A total-bound leaf closes exactly when $\sum_i \ell_i > G$ or $\sum_i u_i < G$. A budget leaf closes when $\ell(S) > b$ or $b + u(S^c) < G$; a target leaf closes when none of its 255 cuts satisfies (4). These are the checker's 77 possible closing reasons. Refinements must tighten at least one bound and leave nonempty coordinate intervals; a split must satisfy $\ell_i \le v < u_i$.

A certificate is a finite tree of refinements and integer splits $x_i \le v$ or $x_i \ge v + 1$. Each leaf is excluded by total bounds or by a cut family with no surviving cut. Induction on this tree proves that its root contains no admissible vector. The separate C++ checker reads these instructions and verifies the deductions; it performs no search and does not use the search program's matching implementation.

Proposition 8 (Certificate replay)**.** *The separate checker accepts an exclusion certificate for each of the 279 projected problems.*

The projected archive contains 499,829 instructions in 313,094 compressed bytes, reducing the instruction count by 95.69% from the original archive. The repository documents both archives and the earlier search computations. Every archived header is checked against its exported root case. Eighty negative controls are rejected, including all 77 possible direct closing claims against a separate $G = 11$ fixture, not a $G = 15$ root problem. The witness satisfies its total, fixed anchor count, coordinate bounds and zero root budget, and fails all 75 targets by physical matching. The fixture is `data/q5-distinct-known-control.txt`; its regeneration and witness checks are in `scripts/check_five_colour_distinct.py`.

Proof of Theorem 1. Lemma 3 gives $N_5 \ge 15$. A solution-free collection of fifteen cubes would satisfy an exported root case by Proposition 6, and would supply a Hall deficiency for every target. Containment and the initial budget tests leave one of 279 roots. Lemma 7 supplies an admissible vector in its projected problem, which Proposition 8 and the sound interval deductions exclude. Hence $N_5 \le 15$. □

6 Formalisation and reproducibility

The package uses Lean 4 and mathlib [7, 8], both at version `v4.35.0`. Lean proves soundness of the reference interval checker, of a faster arithmetic implementation, and of composition of bounded proof chunks. The faster implementation encodes a multiplicity vector as

$$P(x) = \sum_{i=0}^{74} x_i B^i, \qquad B = 1024.$$

For a neighbourhood S, the weight $W(S) = \sum_{i \in S} B^{74-i}$ satisfies

$$x(S) = \left\lfloor \frac{P(x)W(S)}{B^{74}} \right\rfloor \bmod B.$$

Lean proves this identity under $0 \le x_i \le 7$, including the absence of carries; the coefficient bound is $75 \cdot 7 < 1024$. It also proves decoding, coordinate updates and equivalence of the packed feasibility test to the integer inequalities. All 19,125 supplied weights are checked against their neighbourhood masks by the Lean kernel.

Chunk composition requires exact agreement of both boundary vectors and a proof excluding each referenced box; the exporter is untrusted. A limit of 128 instructions gives 4,786 chunks for the projected archive. All 279 projected certificates have passed kernel replay, together with their final assembly into the physical threshold and sharpness theorems. The exact case mapping and certificate hashes are in `data/q5-projected-root-manifest.json`. The 78 declarations listed in `Audit.lean` and the final replay theorems have audited dependencies confined to `propext`, `Classical.choice` and `Quot.sound`; no `native_decide` is used.

Lean proves completeness and uniqueness of the 75 cube varieties and checks all 45,000 compatibility bits using the shared library's palette-independent corner–rotation theorem. The physical lower-bound theorem, `FiveColour.Physical.lower_bound` in `FiveColour/LowerBound.lean`, uses 75 deficient corner sets: 72 singletons and three sets of sizes 5, 6 and 6. In the formal proof, this finite check replaces the elementary target classification in Lemma 3; catalogue completeness and rotation transport extend it to every physical target.

Lean proves the physical Hall reduction, symmetry transport, anchor normalisation, two-variety bounds, anchor selection and the shared-corner argument. Finite checks certify the root capacities and cover representatives. Containment and initial budget tests reduce the 357 roots to 279. Lemma 7 is formalised by `FiveColour.Physical.root_projection` in `FiveColour/RootProjection.lean`; all concrete projection hypotheses and coverage of the retained roots are kernel-checked.

The projected archive's integer semantics are established in `FiveColour/ProjectedBridge.lean`. The identifications include all coordinate bounds, prescribed totals, budgets and Hall-cut families. This checks the entire admissibility predicate, not just the packed boundary vectors. The generated module `FiveColour/ProjectedChunks/Threshold.lean` assembles all 279 exclusions into the unconditional theorems `FiveColour.Packed.ProjectedChunks.threshold15` and `FiveColour.Packed.ProjectedChunks.sharpThreshold15`. The original and intermediate interfaces are documented in `FiveColour/README.md`. Reproduce the full proof and its axiom audit with:

```
make check WORKERS=2
make projected-replay-data WORKERS=8
make projected-replay WORKERS=8
```

The review archive supplies certificates, code and reproduction commands. A Mathlib-only comparator challenge states the puzzle geometrically. Model equivalence is proved by `FiveColour.Puzzle.hasSolution_iff` in `FiveColour/ComparatorBridge.lean`. The solution

module applies the completed threshold theorem and builds in Lean. A successful full external comparator check has not yet been confirmed. Mathematical questions left open here include classifying all fourteen-cube obstructions and determining the five-colour frame numbers for larger cubes.

AI disclosure and review status

GPT-6-Astra (OpenAI), used through Codex, developed the arguments, search and verification programs, Lean formalisation and manuscript under the human author's direction. The author initiated and directed the research. Author review and submission approval of this working manuscript are pending. The inherited geometry has its own provenance in the pinned dependency. Automated checks and AI review are not independent human peer review.

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